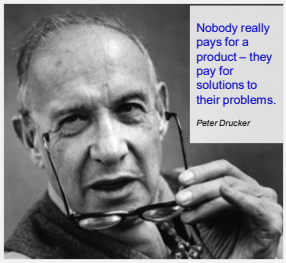
Customer Discovery and Validation



EAS5450: Class #7

1

"Devices" ⇒ "Complete Products"

What we invent **What customers buy**

Technology-Driven ⇒ Market-Driven

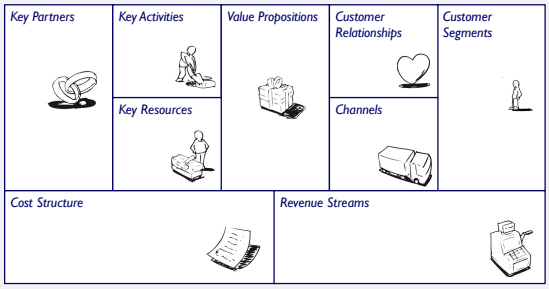
2

Turning our intuitions and enthusiasm into market-driven facts …

3

Recall...

Products Solve Problems

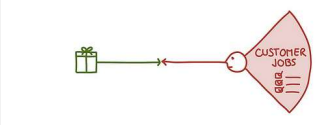


Source: Business Model Generation, Osterwalder, Pigneur et al. www.businessmodelgeneration.com

4

Recall...

The Product **The Problem**

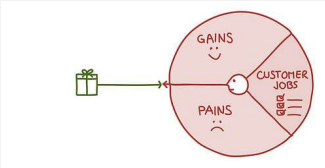


Functional? Social? Emotional? Basic need?

5

Recall...

Benefits customer expects, desires, would be surprised by?



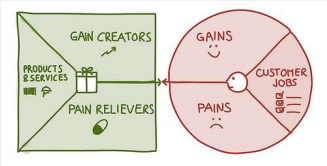
Customer pains before, during, after getting the job done?

6

Recall…

Product-Market Fit

The Product **The Problem**



7

Benefits vs Status Quo

But… the gains promised by a new product must far outweigh the perceived losses from switching away from a familiar solution.

Potential customers over-value losses.

Mismatch between what entrepreneurs think customers want and what customers require to change behavior. Why?

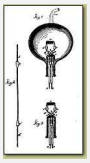
8

A desk is a dangerous place from which to watch the world.

… John le Carré

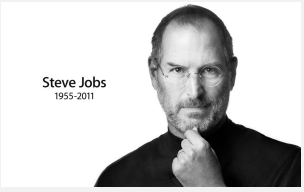
9

"I never perfected an invention that I did not think about in terms of the service it might give to others… I find out what the world needs, then I proceed to invent… "



Thomas Alva Edison
(February 11, 1847 – October 18, 1931)

10



We did design first. We focused on what we thought people would need and want, and how they would interact with their computer. We made sure we got that right, and then we went and figured out how to achieve it technically. We had a design target, and we worked with engineering to reach it.

We applied a lot of creativity and ingenuity on the design side and then pushed the engineers to use the same kind of creativity and innovation to make it happen.

… Cordell Ratzlaff, Mac OSX architect, ca. 1990

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Participatory Design…

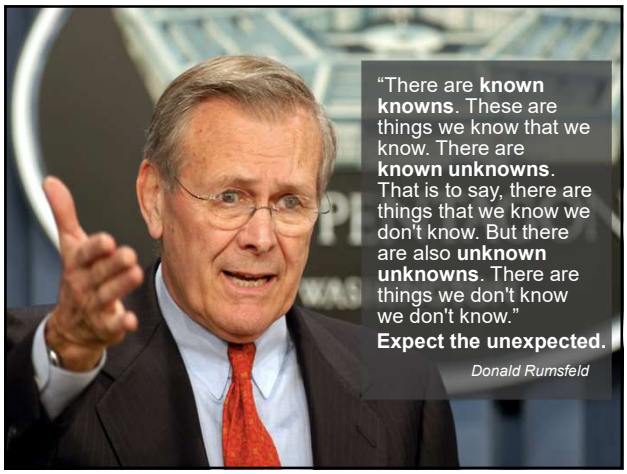
"… it's about what people want to do … you have to know who those people are and what they are really trying to accomplish."

"… it was a lot of trial and error."

"The more user testing…, the smoother it can become. You iterate like that, testing, and then being willing to set aside and build from scratch again."

… Cordell Ratzlaff, Mac OSX architect, ca. 2000

12



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Expect the Unexpected

Recognize False Negatives

"Surprises and reverses can serve as an incentive for great accomplishment. There are no rules here. Just because something doesn't do what you'd planned it to do in the first place doesn't mean it's useless."

"Opportunity is missed by most people because it is dressed in overalls and looks like work."

14

Cognitive Biases

Anchoring on Misleading Perception Impairs Rational Decision-Making

Prof. Daniel Kahneman
Nobel Prize winner

- **Optimism**
  - Reality distortion
- **Planning Fallacy**
  - Tendency to overestimate benefits
  - Tendency to underestimate costs, durations, risks
- **Confirmation Bias**
  - Over-weighting validating results
  - Ignoring contrary results
- **Sunk Costs**
  - "Wasteful" to discard past effort, cost

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Validating Demand

- **Focus Groups**
  - Professional facilitator
  - Open-ended neutral questions
    - Avoid leading the answer
  - Observe body language
  - People try to be nice
    - Don't over-weight "yesses"
    - Negative reaction likely valid
  - Difficult with new products

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Validating Demand

- **Customer surveys**
  - Concept tests
  - Conjoint analyses
  - Perceptual mapping
- **Letters of Intent**
- **β-testing**
  - Usability tests with prototype
  - Market trials with early adopters
  - A/B Split Tests (e.g., MVP Build-Measure-Learn)

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Validating Demand

- **Net Promoter Score (NPS)**

  "On a score of one to ten, how likely is it that you would recommend XYZ to a friend or colleague?"

Score	Class
9 - 10	Promoters
7 - 8	Passives
0 - 6	Detractors

NPS = [% Promoters] – [% Detractors]

- Quick and inexpensive.
- Follow with open-ended question, "Why?"
- Provides early pre-launch indication of product viability and referral rate.
- Provides relative comparison to incumbent rival products.
- Tracked over time post-launch, avoids premature scaling.

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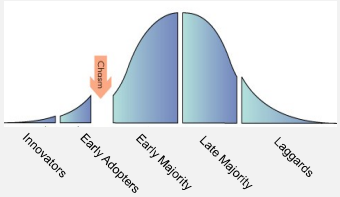
Early Adopter vs Mainstream Psychographics



- G.A.Moore, "Crossing the Chasm"

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Understanding the Roles is Critical

- Initiators
- Gatekeepers
- Influencers
- Deciders
- Purchasers
- Users

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Concept Testing:

Will the Dogs Eat the Dog Food?

EAS5450

Engineering Entrepreneurship I

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A concept is a bundle of product attributes

Attributes:	Concept:
• Connectivity	• "Seamless" docking station
• Size	• Shirt-pocket size
• Weight	• 5.7 ounces
• Battery life	• 8-12 week
• Price	• $299
• Data entry	• User-friendly pen-based

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Concept "A"

$21,100

- ❑ Definitely would buy
- ❑ Probably would buy
- ❑ Might, might not
- ❑ Probably wouldn't
- ❑ Definitely wouldn't buy

Concept "B"

$27,300

- ❑ Definitely would buy
- ❑ Probably would buy
- ❑ Might, might not
- ❑ Probably wouldn't buy
- ❑ Definitely wouldn't buy

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Concept Testing

- Real reactions from real customers
- Primary research
  - Accurate: +/- 10% about 80% of the time
  - Inexpensive
  - Do it yourself

- Improves the likelihood of success.

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Concept Testing…

- Evaluates product viability
- Improves product design
- Enhances product positioning:
  - Segmentation + Differentiation
- Forecasts sales volume = f(price)
- Optimizes pricing:
  - Over-pricing reduces sales
  - Under-pricing leaves $ on the table

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Prior to Concept Test

- Interviews & focus groups
  - Reactions in free-form discussion
    - Narrows target market segments
    - Narrows significant attributes
    - Narrows attribute ranges

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Concept Description

- Mimics how consumer would be exposed to product
- Often resembles advertisement:
  - Presents positioning attribute bundle
  - Pictures and/or text as appropriate
  - Price
- Be consistent – visuals bias results

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Price Testing

- Price affects perception of product
- Expose a respondent to only one price for a given bundle of attributes:
  - For any one respondent, vary price only if at least one other attribute varies
  - Test price points across multiple respondents for a given bundle of attributes.

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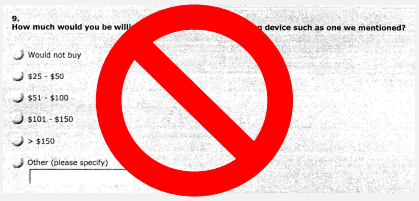
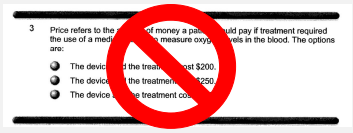
Bad Examples…





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Purchase Intent

- Ask early in test process
- 5-point scale is common:
  - ❑*Definitely would buy*
  - ❑*Probably would buy*
  - ❑*Might or might not*
  - ❑*Probably would not buy*
  - ❑*Definitely would not buy*

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Monadic vs Paired Comparison

- Monadic:
  - In absence of directly competing product
  - When consumers don't shop alternatives
- Paired:
  - When competing product(s) exist
- Monadic followed by paired comparison

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Monadic…

- ❑ Definitely would buy
- ❑ Probably would buy
- ❑ Might, might not
- ❑ Probably wouldn't buy
- ❑ Definitely wouldn't buy

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Paired Comparisons

- Reduces bias by comparing to existing product:
  1. Ask purchase intent for existing product "A"
  2. Ask purchase intent for new product "B"
  3. Market for "B" = difference in "Definitely Would Buy" responses.

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This…

- ❑ Definitely would buy
- ❑ Probably would buy
- ❑ Might, might not
- ❑ Probably wouldn't buy
- ❑ Definitely wouldn't buy

Or this…

- ❑ Definitely would buy
- ❑ Probably would buy
- ❑ Might, might not
- ❑ Probably wouldn't buy
- ❑ Definitely wouldn't buy

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Who to Test

- Target consumer segment(s)
  - Concept test can help define segment(s)
- All influencers of purchase decision
- Other stakeholders (esp. channel intermediaries)

35

Where to Test

- Malls, train stations, etc.
- By phone (if visuals not important)
- By mail, email or web-based surveys
- Dummy website:
  - Simulate ordering process
  - Ethical questions: reveal simulation promptly

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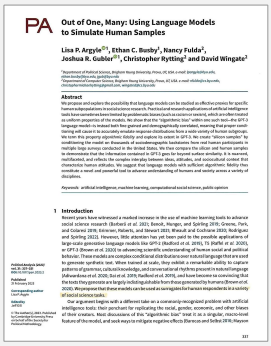
Using LLMs to Simulate Human Surveys



PA

Out of One, Many: Using Language Models to Simulate Human Samples

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Abstract

We propose and explore the possibility that language models can be studied as effective proxies for specific human subpopulations in social science research. Practical and research applications of artificial intelligence tools have sometimes been limited by problematic biases (such as racism or sexism), which are oftentimes attributed as uniform properties of the models. We show that the "algorithmic bias" within one such tool—the GPT-3 language model—is instead both highly conditioned by subtle variations in input prompts and sufficiently flexible to enable "silicon samples" to accurately emulate response distributions from a wide variety of human subgroups. We term this property algorithmic fidelity and explore its extent in GPT-3. We create "silicon samples" by conditioning the model on thousands of sociodemographic backstories from real human participants in multiple large surveys conducted in the United States. We then compare the silicon and human samples to demonstrate that the language model GPT-3 is capable of emulating organizations of humans that are multifaceted, and reflect the complex interplay between ideas, attitudes, and sociocultural context that characterize human attitudes. We suggest that language models with sufficient algorithmic fidelity thus constitute a novel and powerful tool to advance understanding of humans and society across a variety of disciplines.

Keywords: artificial intelligence; machine learning; computational social sciences; public opinion

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1 **Introduction**

Recent years have witnessed a marked increase in the use of machine learning tools to advance social science research (Bushell et al. 2021; Benoit, Hungar, and Spirling 2019; Grimmer, Park, and Colaresi 2019; Grimmer, Roberts, and Stewart 2021; Wesslett and Cochrane 2019; Rodriguez and Spirling 2022; Ziems et al. 2023). Much of this work has been enabled by the emergence of large-scale generative language models like GPT-3 (Radford et al. 2019), T5 (Raffel et al. 2020), or GPT-3 (Brown et al. 2020) in advancing scientific understanding of human social and political behavior. These models are complex conditional distributions over natural language that are used to generate synthetic text. When trained at scale, they exhibit a remarkable ability to capture patterns of grammar, cultural knowledge, and conversational rhythms present in natural language (Adiwardana et al. 2020; Zut et al. 2019; Radford et al. 2019), and have become so convincing that the texts they generate are largely indistinguishable from those generated by humans (Brown et al. 2020). We propose that these models can be used as surrogates for human respondents in a variety of social science tasks.

Our argument begins with a different take on a conversely-recognized problem with artificial intelligence tools their penchant for replicating the racial, gender, economic, and other biases of their creators. Most discussions of this "algorithmic bias" treat it as a singular, macro-level feature of the models, and seek ways to mitigate negative effects (Barocas and Selbst 2019; Mayson

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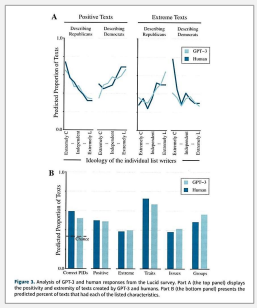

Figure 3. Analysis of GPT-3 and human responses from the Lucid survey. Part A (the top panel) displays the positivity and extremity of texts created by GPT-3 and humans. Part B (the bottom panel) presents the predicted percent of texts that had each of the listed characteristics.

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Analyzing Results

- Respondents try to be nice
- Want 80-90% in top 2 boxes
- Responses below top box are risky:
  - Top box is good indicator of likely trial rate
  - May get some (< half) of second box
- Predicts initial trial rate only

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Follow-Up Questions

- How well is the concept understood?
- Uniqueness vs other products
- Level of believability
- Importance for solving problem(s)
- Inherent interest
- Value for money (5-point scale)
- Purchase intent driver(s)

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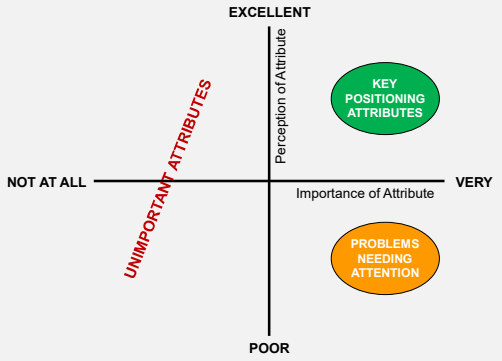
Follow-Up Quadrant Analysis



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More Follow-Up Questions

- Respondent profile
  - Who else influences decision?
  - Demographics
  - Psychographics
  - Perception of overall product category
  - Barriers to switching:
    - Cost: time and/or money
    - Loyalty
    - Satisfaction

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Market Share Factors

- Market share=
  - % of top box responses x
  - % of target segment aware of product x
  - % of those aware who understand benefits x
  - % in segment who can easily purchase product.
- Run sensitivity tests around these factors

Important!

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Conjoint Analysis
& Perceptual Mapping

43

- Concept Testing: Reaction (purchase intent) reflects underlying preferences.
- Conjoint Analysis: Provides insight about underlying preferences.

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Conjoint Analysis

"One of the most celebrated tools in marketing"

- Prof. Eric Bradlow

E-ZPass

- 1000's performed annually.
- Software available (www.sawtoothsoftware.com).
- More expensive than Concept Testing.
- Constructs customers' value systems.

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Conjoint Demonstration

Analysis: Fast Cars				
Speed	80 MPH	120 MPH	155 MPH	200 MPH
Price	$40K	$80K	$120K	$200K
Color	Red	Black	Silver	

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Choice 1 of 15…

- ❑ 80 mph / black / $80,000
- ■ 200 mph / red / $120,000
- ❑ 120 mph/ red / $80,000
- ❑ None of these

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Part-Worth Utilities

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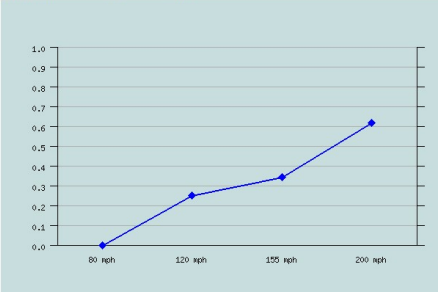
Speed

level	display count	chosen count	part-worth utility
80 mph	31	0	0.00
120 mph	36	9	0.25
155 mph	44	15	0.34
200 mph	39	24	0.62



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Color

level	display count	chosen count	part-worth utility
Red	45	17	0.38
Black	52	17	0.33
Silver	53	14	0.26

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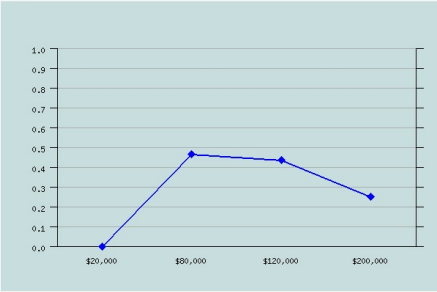
Price

level	display count	chosen count	part-worth utility
$20,000	29	0	0.00
$80,000	43	20	0.47
$120,000	46	20	0.43
$200,000	32	8	0.25



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Attribute Importance

Importance of Attribute X = Range of part-worth utility (X) divided by the sum of part-worth utility ranges of all attributes.

Attribute	u Max.	u Min.	Range	Importance
Speed	0.62	0	0.62	**51%**
Price	0.47	0	0.47	**39%**
Color	0.38	0.26	0.12	**10%**
Sum of Ranges			**=1.21**	

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Additive Conjoint Model

$$U_i = \sum_{j=1}^{M} u_j(x_{ij})$$

U_i : Utility (preference) for Concept "i"
M : Number of attributes
u_j : Part-worth utility of attribute "j"
X_{ij} : Level of attribute "j" for Concept "i"

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Utility of Concept "A"

$$U_A = u_{color}(color_A) + u_{speed}(speed_A) + u_{price}(price_A)$$

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Estimating Market Share

Market share for Concept N = Utility(N) divided by the sum of the utilities of all available concepts including N.

	Concept A		Concept B		Concept C		
Speed	200	.62	120	.25	80	0	
Price	$120K	.43	$80K	.47	$20K	0	
Color	Red	.38	Silver	.26	Black	.33	
Utility		1.43		0.98		0.33	=2.74
Share		52%		36%		12%	

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Multiattribute Utility Model

(3-Attribute Example)

$$U = k_1u_S + k_2u_P + k_3u_C + k_4u_Su_P + k_5u_Su_C + k_6u_Pu_C + k_7u_Su_Pu_C$$

U_i : Utility of Concept "i"
k_n : Constants found by data regression
u_j : Part-worth utility of attribute "j" for Concept "i"

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Multiattribute Utility Model

(Simplified)

$$U_i = k_1u_{S,i} + k_2u_{P,i} + k_3u_{C,i}$$

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Market Share with Weighted Utilities

Market share for Concept N = Weighted utility(N) divided by the sum of the weighted utilities of all Concepts.

	Concept A		Concept B		Concept C		
Speed k=0.51	200	.316	120	.127	80	0	
Price k=0.39	$120K	.167	$80K	.183	$20K	0	
Color k=0.10	Red	.038	Silver	.026	Black	.033	
Utility		0.521		0.336		0.033	=0.89
Share		58%		38%		4%	

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Steps…

1. Select relevant (actionable) attributes.
2. Choose relevant levels for each attribute.
3. Select a response method.
4. Define market segment & sample size.
5. Administer survey.
6. Analyze data and compute customer value systems.
7. Provide results.

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Conjoint Analysis Applications

- New Product Development
  - Concept optimization
  - Product differentiation
- Market Segmentation
- Price Elasticity Investigation
- Market Share Analysis

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Online Survey Sites

- Sawtooth Software
- Survey Analytics
- Zoomerang
- Survey Monkey
- Qualtrics
- Comscore

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Perceptual Mapping

"Profile Analysis"

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Preference -vs- Perception

- **Preference** – what people like.
- **Perception** – what people see.
- **Perceptual Mapping** – evaluates relative product positioning, as seen by consumers, of competing products.

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vs.

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Profile Analysis Data

Average scores of 100 respondents	Score on a 1-10 scale 1=Completely Disagree 10=Completely Agree	
	PRITT	UHU
Conveys confidence	6.8	8.5
Is of high quality	7.2	8.4
Is for special uses only	6.5	5.5
Is only for sticking paper	5.7	4.3
Is somehow likeable	6.4	7.2
Glues reliably	7.1	8.4
Is of great strength	6.9	8.3
Is a well-known brand	7.9	9.7
Is for general purposes	6.5	8.1
Is usually reasonably price	6.5	6.7
Is sold everywhere	7.4	8.9
Offers all adhesives you need	7.1	8.3
Is more for children	4.9	4.5
Is more known from the office	6.3	5.6
Offers useful adhesives	8.0	8.7
Has a well-structured range	6.9	7.9
Is something for experts	2.4	3.0
Is clean in use	7.6	7.7

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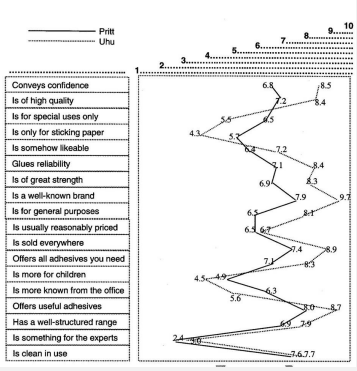
"Snake Plot" of Relative Positions



- **UHU Brand**
  - Stronger
  - General purpose
- **PRITT Brand**
  - For sticking paper
  - Special uses

From: *Analyzing Consumer Perceptions*, Harvard Business School Press, Case #9-599-110 (1999)

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“Devices” ⇒ “Complete Products”

What we invent **What customers buy**

Turning our intuitions and enthusiasm into market-driven facts.

67